Journal of Applied Developmental Psychology 30 (2009) 310–320

Contents lists available at ScienceDirect

Journal of Applied Developmental Psychology

The contribution of inhibitory control to preschoolers' social–emotional competence☆

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ARTICLE INFO

Available online 23 February 2009

Keywords:
Executive function
Inhibitory control
Social–emotional competence
Preschoolers

ABSTRACT

Social–emotional competence is a key developmental task during early childhood. This study examined concurrent relationships between maternal education and employment status, children's sex, ethnicity, age, receptive vocabulary, emotional knowledge, attention skills, inhibitory control and social–emotional competence in a sample of 146 preschool, low-income, ethnically diverse children from Head Start classrooms. Multilevel models demonstrated that inhibitory control played a significant role in the concurrent prediction of teacher ratings of social–emotional competence above and beyond other variables associated with social–emotional competence. Children who demonstrated better inhibitory control were more likely to be rated higher on social skills and lower in internalizing behaviors. Findings suggest that early identification of inhibitory control difficulties may be beneficial for targeting children at risk for maladaptive outcomes. The contribution of environmental experience to the development of inhibitory control skills suggests there are many opportunities to intervene during early childhood.

1. Introduction

The development of social–emotional competence is a process of acquiring a set of interrelated skills that promote emotion recognition and management, compassion for others, positive social relationships, and responsible and effective decision-making (Denham & Weissberg, 2004; Greenberg, Weissberg, O'Brien, Zins, Fredericks, Resnik, 2003). The ability to manage behaviors and emotions in accordance with societal expectations is an essential component of social-competence that is especially important during early childhood as it has been shown to play an important role in school adjustment and achievement during kindergarten (Howse, Calkins, Anastopoulos, Keane, & Shelton, 2003; Shields, Dickstein, Seifer, Giusti, & Spritz, 2001). There has been considerable theorizing and some data indicating that executive function, and in particular inhibitory control, may facilitate this process (Hughes, Dunn, & White, 1998; Kochanska, Coy, & Murray, 2001; Rothbart, Ahadi, & Hershey, 1994). Inhibitory control, the cognitive-related ability to inhibit a strong dominant response in favor of a subdominant one, could serve as a key internal resource for young children faced with the social challenges presented by a preschool classroom.

To better understand the role inhibitory control plays in preschool children's development of social–emotional competence, the present study used an at-risk, low-income sample of preschoolers to examine the concurrent relationships between inhibitory control abilities and teacher-rated social skills, internalizing and externalizing problems above and beyond other factors known to be associated with these behaviors.

☆ Support for this research was provided by Head Start/University Partnership Grant (#90YD0063) and the National Institute on Drug Abuse (DA 017629).
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0193-3973/$ – see front matter
doi:10.1016/j.appdev.2008.12.012

1.1. Inhibitory control in early childhood

Preschool represents an important developmental period to study individual differences in inhibitory control and their associations with behavioral outcomes. Inhibitory control emerges in late infancy and exhibits dramatic development in the preschool years (Diamond, 2002; Rothbart, 1989). Research consistently shows that young children have difficulty inhibiting impulses (Diamond, Kirkham, & Amso, 2002; Diamond & Taylor, 1996; Dowsett & Livesey, 2000; Gerstadt, Hong, & Diamond, 1994). Although age-related improvements in inhibitory control occur from three to six years, children less than five years old still have substantial difficulty inhibiting a prepotent response (Gerstadt et al., 1994).

Inhibitory control has been conceptualized in a variety of ways throughout the early childhood literature (see Kochanska, Murray, & Harlan, 2000). Effortful control was first introduced by Rothbart and colleagues as the self-regulatory aspect of temperament that enables children to regulate their anger and approach systems (Rothbart, 1989; Rothbart & Bates, 1998). Effortful control is part of the active inhibitory system and has been described as the ability to inhibit a prepotent or dominant response in favor of a subdominant one (Kochanska et al., 2000; Olson, Sameroff, Kerr, Lopez, & Wellman, 2005). It is believed that effortful control allows children to regulate their actions in the service of current and future goals (Rueda, Posner, & Rothbart, 2005). Effortful control first emerges in the second half of the first year of life with individual differences becoming increasingly stable during toddlerhood and into preschool (Kochanska et al., 2000).

Other researchers suggest that inhibitory control is one aspect of the multi-component construct of executive function (EF). EF contributes to the ability to coordinate thoughts and actions in a goal-directed manner (Kopp, 1989; Perner & Lang, 1999) and includes planning, working memory, and attentional control skills in addition to inhibitory control (Hughes, 1998; Pennington, 1997; Welsh, Pennington, & Groisser, 1991). Rudimentary forms of EF emerge in the first year of life and continue to develop until at least early adulthood (Dempster & Corkhill, 1999; Diamond, 1991; Welsh & Pennington, 1988). EF research suggests that inhibitory control is especially relevant to the development of children's abilities to regulate their behaviors and emotions (Carlson, Moses, & Brenton, 2002; Hughes et al., 1998; Hughes, White, Sharpen & Dunn, 2000). However, it is important to note that because EF tasks are inherently complex, no pure measure of any single EF component exists. Most EF tasks (including the ones used in the present study) tap into multiple components (Diamond, Kirkham, & Amso, 2002; Pennington, 1997).

Olson and colleagues posit that inhibitory control is one facet of the broader multi-dimensional construct of impulse control (Olson, 1989; Olson, Schilling, & Bates, 1999). Their findings suggest that impulse control is composed of a cognitive dimension and a delay dimension. Inhibitory control falls under the cognitive dimension and is generally measured using highly structured tasks where children are required to inhibit action in a non-emotional task. On the other hand, the delay dimension of impulse control has been measured using tasks like "delay of gratification" or "resistance to temptation" where children are required to comply with social expectations of appropriate behavior in a more emotionally-eliciting situation (Olson, 1989). Some research indicates that the delay dimension of impulse control may be a stronger predictor of children's outcomes than the cognitive dimension (Olson, 1989; Olson & Hoza, 1993; Olson et al., 1999). However, the cognitive dimension remains a significant aspect of impulse control, which may influence children's classroom behaviors.

Children's ability to inhibit impulses plays a crucial role in everyday peer and adult interactions in the preschool classroom. For instance, preschool children are likely to encounter peer conflict in situations when other children obstruct or interfere with their goals (e.g., another child takes away the toy he/she was playing with/takes his/her place in line etc.). For many children, their natural tendency may be to use physical aggression (e.g., hit the child who took his/her toy away, push the child who took his/her space in line) to attain their goals. However, children who are better able to inhibit their natural tendency to use physical aggression (i.e., dominant response) and instead use their words (i.e., subdominant response) to attain their goals are likely to be viewed by their peers and teachers as better play partners and therefore more socially competent. On the other hand, those children who are unable to inhibit their impulses are more likely to exhibit externalizing behaviors such as physical aggression and to be viewed by their peers and teachers as poor play partners. They may be rejected from play groups, which may lead to internalizing behaviors.

1.2. The relationship between inhibitory control and behavior

Several studies have investigated the relationship between inhibitory control skills and behavioral outcomes in middle-class children. Kochanska and colleagues reported that effortful/inhibitory control in early childhood was longitudinally related to better emotion regulation, fewer externalizing problems, stronger conscience, and greater committed compliance later in childhood (Kochanska & Knaack, 2003; Kochanska et al., 2000, 2001). Olson et al. (2005) reported that children's effortful control at age three was a significant negative predictor of externalizing behavior above and beyond IQ, dispositional anger, and psychosocial adversity in a middle-class sample. In another study, Olson et al. (1999) found that laboratory measures of both cognitive and delay dimensions of inhibitory control at six and eight years old predicted parent and self-ratings of externalizing behavior problems during adolescence.

Research also has examined inhibitory control in young children at risk for developing ADHD (Campbell, Pierce, March, Ewing, & Szumowski, 1994; Hughes et al., 1998, 2000; Perner, Winfried, & Barchfeld, 2002). Hughes and colleagues found that although preschoolers with high rates of hyperactive symptoms were more likely to fail a variety of EF tasks (i.e., planning, working memory, inhibition, attentional flexibility), inhibitory control was the only EF dimension that remained significant after controlling for verbal skill, age, and family background (Hughes et al., 1998). Additionally, inhibitory control skills were negatively related to antisocial behavior and positively related to prosocial behaviors during peer interactions (Hughes et al., 2000). Hughes and colleagues consistently report that inhibitory control is the most important skill predicting young children's behavior.

Others have examined EF, broadly defined, in children with elevated problem behaviors (Cole, Usher, & Cargo, 1993; Kusche, Cook, & Greenberg, 1993; Seguin & Zelazo, 2005; Speltz, DeKlyen, Calderon, Greenberg, & Fisher, 1999). The majority of these studies created composite EF scores by aggregating across various tasks, thereby overlooking the significant role of inhibitory control. They found that children exhibiting behavior problems had poorer EF skills in general compared to normally developing children. Within an intervention context, Riggs, Greenberg, Kuche, & Pentz (2006) found that children exposed to a social–emotional intervention had fewer problem behaviors at one year post-intervention as compared to control children and that this intervention effect was mediated by improvements in inhibitory control.

Although the aforementioned research, largely conducted on middle class samples, suggests that inhibitory control is central to the development of social–emotional competence, it is important to investigate whether this relationship holds for low-income children as well in order to understand the generalizability of these findings. Very few reports have examined inhibitory control in samples of disadvantaged children (Blair, 2003; Blair & Peters, 2003). In a sample of Head Start children, Blair (2003) and Blair and Peters (2003), reported a positive association between teachers' reports of children's on task behavior in the classroom and high levels of inhibitory control as assessed by a composite score of the Peg Tapping and Day/Night tasks. Although this provides some support for the generalization of this relationship to disadvantaged population, the present study aims to provide more evidence by examining these relationships in a larger and more ethnically diverse sample of Head Start children.

1.3. Predictors of social–emotional competence in preschool

The above findings suggest that children who are better able to inhibit their behavior have better social skills and fewer behavior problems. However, there has been little research that has integrated the assessment of inhibitory control in preschoolers with the examination of other factors that have also been shown to predict social and emotional competence. Children's sex (Lavigne, Gibbons, Christoffel, Arend, & Rosenbaum, 1995; Spieker, Larson, Lewis, Keller, & Gilchrist, 1999), academic skills (Malecki & Elliott, 2002), emotional knowledge (Arsenio, Cooperman, & Lover, 2000; Denham, Blair, DeMulderr, Levitas, & Sawyer, 2003; Fine, Izard, Mostow, Trentacosta, & Ackerman, 2003) and attention skills (Belsky, Friedman, & Hsieh, 2001; Eisenberg, Fabes, Nyman, Bernzweig, & Pinuelas, 1994; Raver, Blackburn, Bancroft, & Torp, 1999) have been identified as important predictors of social–emotional competence. Caregivers' education and employment status have also been shown to predict children's social–emotional skills (Bradley & Corwyn, 2002). There has been little examination of how inhibitory control may interact with these predictors or how it may relate to social competence and behavior problems in young children above and beyond the impact of the aforementioned factors.

1.4. The present study

This study contributes to the field in three ways. First, it examined the relationship between inhibitory control and social–emotional competence in a disadvantaged sample of ethnically-diverse, urban preschoolers. Second, by using multi-level modeling it examined whether classroom-level effects influence relations between factors and accurately accounted for the effect of classroom-level effects on individual-level relations. Third, the present study controlled for factors already shown to influence social–emotional competence in order to isolate the impact of inhibitory control. To this aim, two primary research questions were addressed:

1) What is the role of inhibitory control in predicting children's social–emotional competence, controlling for factors that have already been shown to predict social skills and behavior problems (maternal education and employment status, children's sex, ethnicity, age, receptive vocabulary, emotional knowledge, attention)?
2) Does the relationship between inhibitory control and social–emotional competence vary by maternal education and employment status, children's sex, ethnicity, receptive vocabulary, emotional knowledge and/or attention skills?

It is hypothesized that children with better inhibitory control would have higher social skills and lower externalizing and internalizing ratings, above and beyond the impact of the aforementioned factors. Because of the limited research on what factors might moderate the relationship between inhibitory control and social–emotional competence, we did not pose any a priori hypotheses regarding the moderation analyses.

2. Method

2.1. Design

The present study examined behavior of a subset of preschool children from a randomized clinical trial of the PATHS (*P*romoting *A*lternative *TH*inking *S*trategies) preschool program, a curriculum developed to promote social and emotional competence (Domitrovich, Greenberg, Cortes, & Kusche, 2005). Time 1 data for both control and intervention children were used in the present analyses. Time 1 assessments were conducted at the beginning of the school year during the fall of 2000. As these data were collected at the pretest/Time 1, no intervention had been initiated (see Domitrovich, Cortes, & Greenberg, 2007 for results from the impact study).

2.2. Participants

There were 246 children ranging in age from 2.5 to 5 years old in the PATHS program at Time 1. However, only children four and older were included because of the need for valid assessment of the present study's constructs. This study therefore included 146 children who were 4–5 years old. See Table 1 for a description of the sample participants and their caregivers.

2.3. Measures

2.3.1. Receptive vocabulary

The Peabody Picture Vocabulary Test-Revised (PPVT-R; Dunn & Dunn, 1981) is a well-established measure of children's receptive vocabulary. The interviewer presented the child with four pictures and asked him/her to identify the picture of the word that was orally presented to him/her. The measure consists of 175 items. The PPVT-R shows good convergent validity, and the split-half reliability coefficient for the PPVT-R is .71 for children 48 to 53 months of age. The median internal consistency across age groups is $r = .95$ (Dunn & Dunn, 1981).

2.3.2. Emotional knowledge

The Kusche Emotional Inventory (KEI, Kusche, 1984) was developed to assess children's recognition of various emotion labels (e.g., happy, sad, mad, scared, love). A version of the Recognition of Emotion Concepts subtest, adapted for 4-year olds, was used in this study (Speltz et al., 1999). Children were presented with a page divided into four sections with four cartoon faces, each with a particular emotional expression. Out of the four cartoons, there was one target picture and three distracter pictures. Children were presented with 30 stimuli pages assessing 15 different emotions and asked to identify the cartoon picture that best matched the emotion stated. For each trial, children received two points for the correct response and zero for an incorrect response. If children chose a picture that depicted an emotional expression of the same valence (e.g., child answered sad for a mad expression) as the target feeling word, they received one point. The highest possible score was 60. For the original version of this subtest, Kusche (in press) reported a split-half reliability of .89 and test–retest reliability of .85. In the present sample, there was adequate internal consistency with a Cronbach's alpha coefficient of .73. Total percent correct was used as the predictor variable in the present analyses.

2.3.3. Attention skills

The Leiter-Revised Attention Sustained Subtest (Leiter-R AS; Roid & Miller, 1997) assessed children's ability to sustain attention to detail in a repetitive task. Children were shown a target figure (e.g., flower) located at the top of the stimulus and were instructed to scan an array of figures and cross out all of the target figures as quickly as they can. Children were administered a series of four boxes, each with a different target figure located in an increasingly more complicated array of figures. For each box, they were given up to 30 s to complete the task. An adjusted total correct score was created by subtracting the total number of errors from the total number of correct. This score was converted into an adjusted total scaled score where the average score is represented by a scaled score of 10 ($SD = 3$). The Leiter-R AS has been shown to have good internal consistency with a Cronbach's alpha coefficient of .83 for the 4–5 year old version and good test–retest reliability of ($r = .85$).

Table 1
Descriptive statistics for participants and caregivers ($N = 146$).

	Mean (*SD*) or percent
Participant characteristic	
Age	54 months (3.6)
Female	54%
Ethnicity	
African-American	49.3%
Anglo-American	38.4%
Hispanic	6.8%
Multi-racial or other	5.5%
Caregiver characteristic	
Age	32 years (9.8)
Income	$6642 (6320)
Relationship to child	
Biological mother	81.3%
Relative, foster parent or guardian	18.3%
Employment	
Full or part-time employed	60.3%
Unemployed	39.7%
Education	
Some high school	22.8%
GED or high school diploma	50.4%
Some college	12.3%
Associates or bachelor degree	14.2%

2.3.4. Inhibitory control

Two tasks were used to assess children's inhibitory control skills. One measure was a Stroop-like task called Day/Night (D/N; Gerstadt et al., 1994). This measure assesses children's ability to inhibit a natural tendency to give a verbal response while remembering the rule for the correct response. Children were presented with a series of two types of cards. One was a black card with stars and one was a white card with a bright sun. Children were instructed to say “day” to the black cards and “night” to the white cards. Children were given the opportunity to practice each action following the experimenter's instructions. Then, children were given up to three practice trials and praised for correct responses and corrected for incorrect responses. If children got any trials correct during the practice trials, they were included as their first two trials. In all, children were administered 16 trials in which eight of each type of card was included in a counterbalanced sequence. The internal consistency was good within the present sample with a Kuder–Richardson (*KR-20*) coefficient of .89.

The other measure of inhibitory control was Luria's (1966) tapping task called Peg Tapping (PT; Diamond & Taylor, 1996). This measure was administered to assess children's ability to inhibit a natural tendency to mimic an action while remembering the rule for the correct action. The administration procedures described by Diamond and her colleagues were followed. Children were instructed to tap a wooden dowel (15-cm length × 1-cm diameter) twice when the experimenter tapped once and to tap once when the experimenter tapped twice. The procedures for administration of the PT task were similar to those described above for the D/N task. In all, children were administered 16 trials in which eight of each action (one-tap and two-tap) were included in a counterbalanced sequence. The internal consistency was good within the present sample with a *KR-20* of .88. Although both tasks are well-established measures of inhibitory control, it is important to recognize that neither is solely a measure of inhibitory control, but also requires working memory. For the purpose of the present study, however, these tasks will be referred to as inhibitory control tasks.

Similar to previous work with these two measures, the scores on D/N and PT were moderately correlated ($r = .42$, $p < .001$). Although previous studies have created a single indicator of inhibitory control by averaging the total percent correct on both D/N and PT (e.g., Blair, 2003; Blair & Peters, 2003), the present study analyzed scores for each measure separately in order to examine their independent contributions in the prediction of children's social–emotional competence.

2.3.5. Social–emotional competence

The Preschool and Kindergarten Behavior Scales (PKBS; Merrell, 1996), a teacher report measure, was designed to assess social skills and problem behaviors during the past month in 3–6 year old children. The Social Skills scale consisted of 34 items that describe adaptive or positive behaviors associated with positive personal and social outcomes (e.g., is cooperative, follows directions, shows self-control). These items were designed to reflect both peer-related and adult-related forms of social adjustment. The Problem Behavior scale consists of 42 items that were designed to reflect common preschool and kindergarten children behavior problems (e.g., has tantrums, will not share, is physically aggressive). The two subscales, externalizing and internalizing behaviors, were used for the present study. For all items, responses were based on a four-point scale: 0 = never true, 1 = rarely true, 2 = sometimes true, and 3 = often true. The subscales showed good internal consistency, with Cronbach's alpha coefficients ranging from .84 to .97 in the present sample. The present study used mean scores for each scale as the outcome variables.

2.4. Procedure

Children were tested outside the classroom and completed the battery of measures in a quiet area of the Head Start center during two separate, 30-minute sessions within a two-month period. The first session included the Peabody Picture Vocabulary Task-Revised and the Kusche Emotional Inventory. The second session included the Leiter-R Sustained Attention Task, Day/Night Task and Peg Tapping. Teacher ratings of children's behaviors were collected at this time as well.

Demographic information was collected from the primary caregiver at the first home visit with the child's teacher. It included questions about the caregiver's and child's ethnicity, age, and the caregiver's relationship to the child, his or her education, employment status, current income, and marital status.

3. Results

3.1. Missing data treatment

Out of the total sample ($N = 146$), 25 children were missing at least one child-based assessment: one child was missing only the PPVT-R; three children were missing only the Leiter-R AS; five children were missing only the D/N measure; six children were missing only the PT measure; one child was missing both the KEI and the Leiter-R AS measures; one child was missing both the Leiter-R AS and the PT measures; and eight children were missing both the D/N and PT measures. Also, two children did not receive teacher ratings at Time 1. Reasons for missing data include interviewer error, child refusal to complete the measure, and child's inability to understand the instructions. To effectively use all available data, multiple imputation was used to calculate the parameter estimates. For the present analyses, five datasets (Schafer & Olsen, 1998) were imputed using PROC MI in SAS v9.1. PROC MI Analyze was used to combine parameter estimates from the imputed datasets (Schafer & Graham, 2002). Although the parameter estimates were very similar using the multiple imputation procedure compared to listwise deletion, the multiple imputation analyses increases power to detect significant differences and corrects for any potential selection bias for children who

Table 2
Mean (and *SD*) child measure scores and teacher-rated outcomes.

Measure	Mean (SD)
PPVT standard score	89.61 (12.81)
KEI % correct	65.11 (13.05)
Leiter-R AS adjusted correct scaled score	10.75 (2.90)
D/N % correct	78.38 (24.88)
PT % correct	68.21 (29.78)
PKBS social skills	2.35 (.48)
PKBS externalizing	.75 (.60)
PKBS internalizing	.64 (.60)

completed the measures compared to those who did not. Therefore, the values obtained using PROC MI Analyze will be reported in the following sections.

3.2. Preliminary analyses

Table 2 presents descriptive statistics for the child and teacher report variables. In this low-income sample, children scored close to one standard deviation below the national norm for receptive vocabulary. However, they demonstrated average sustained attention with scores comparable with the national norm. On average, they achieved a somewhat greater percent of items correct on the Day/Night measure of inhibitory control compared to scores on the Peg Tapping measure. In general, the children in this sample were rated by teachers as having good social skills and low levels of externalizing and internalizing behaviors at the beginning of the school year.

Table 3 presents the correlations among the child variables. Receptive vocabulary was strongly related to emotional knowledge and moderately related to sustained attention. There was a significant positive relation between the inhibitory control, as measured by PT, and both emotional knowledge and sustained attention. Emotional knowledge was moderately, positively correlated with sustained attention.

Table 3 also presents correlations between the child constructs and the three teacher-rated subscales. There were no significant relationships between sustained attention and social skills or problem behaviors. Inhibitory control, as measured by both D/N and PT, was positively correlated with social skills. However, only PT was significantly (negatively) correlated with externalizing and internalizing problems. Both receptive vocabulary and emotional knowledge were related to children's social skills, indicating these scores should be covaried in order to examine the role of inhibitory control.

3.3. Analytic method for the multi-level models

Because children in this sample were nested within classrooms, multilevel models with child-level variables at level 1 and classroom membership at level 2 were used to examine the relationship between inhibitory control and teacher-rated social–emotional competence. By specifying a 2-level model we were able to control for any interdependence in teacher ratings due to children being in the same classroom and having the same rater. The intraclass correlation (ICC) from the empty model (i.e., no predictors included) was .16 for social skills, .23 for externalizing behavior, and .37 for internalizing behaviors. These numbers represent the correlation of teacher ratings for children in the same classroom.

For the 2-level models predicting each of the three outcomes, child-level variables were created as predictors. The child-level variables were grand-mean centered and represent between-person differences. For instance, the child-level PPVT-R variable (i.e., receptive vocabulary) represents the deviation between the overall mean PPVT-R score (90) and the child's individual PPVT-R score. SAS PROC MIXED was used to estimate linear mixed models to assess the between-person differences in maternal education and employment status, children's sex, ethnicity, age, receptive vocabulary, emotional knowledge, attention skills, and inhibitory

Table 3
Concurrent correlations among child constructs and between child constructs and social–emotional competence.

	Receptive vocabulary	Emotional knowledge	Sustained attention	IC: day/night	IC: peg tapping
Receptive vocabulary					
Emotional knowledge	.52***				
Sustained attention	.25**	.21*			
IC: day night	.03	.14	.01		
IC: peg tapping	.15	.29**	.26**	.42***	
Social skills	.38***	.35***	.13	.19*	.34**
Externalizing	− .19*	− .11	− .13	− .08	− .22*
Internalizing	− .16	− .05	− .10	− .15	− .35***

*$p < .05$, **$p < .01$, ***$p < .001$.

Table 4
Parameter estimates for multi-level model predicting social skills, externalizing problems and internalizing problems.

Fixed effects	Social skills			Externalizing problems			Internalizing problems		
	Estimate	*SE*	*p* value	Estimate	*SE*	*p* value	Estimate	*SE*	*p* value
Intercept	**2.274**	0.083	< .0001	**0.699**	0.138	< .0001	**0.588**	0.115	< .0001
Maternal education	− 0.042	0.056	0.4487	0.096	0.082	0.2457	0.007	0.059	0.9106
Maternal employment status	0.013	0.045	0.7799	− 0.087	0.065	0.1847	− 0.041	0.048	0.3901
Sex	0.077	0.073	0.2897	0.028	0.104	0.7884	0.053	0.077	0.4950
Ethnicity	0.125	0.083	0.1306	− 0.027	0.122	0.8235	0.037	0.092	0.6847
Age	0.035	0.129	0.7842	− 0.011	0.183	0.9525	− 0.014	0.136	0.9158
Receptive vocabulary	**0.009**	0.003	0.0106	**− 0.010**	0.005	0.0361	**− 0.010**	0.004	0.0101
Emotional knowledge	0.005	0.003	0.1667	− 0.002	0.005	0.6676	0.003	0.004	0.4405
Sustained attention	− 0.009	0.015	0.5469	− 0.002	0.020	0.9311	0.014	0.014	0.3141
Inhibitory control: D/N	0.001	0.002	0.4156	− 0.000	0.003	0.8960	− 0.000	0.002	0.9695
Inhibitory control: PT	**0.005**	0.002	0.0240	− 0.004	0.003	0.2321	**− 0.006**	0.002	0.0008

Note. Bolded parameter estimates are significant at $p < .05$.

control in the prediction of social skills, externalizing and internalizing behavior problems. For each model, fixed effects of the predictor variables were added to the empty 2-level model. Model parameters are reported in Table 4.

3.4. The role of inhibitory control in prediction of social–emotional competence

3.4.1. Social skills

Parameter estimates for this model can be found on the left panel of Table 4. The fixed effects of receptive vocabulary ($p < .05$) and inhibitory control (as measured by PT; $p < .01$) were significant. The parameter estimates suggest that children with greater receptive vocabulary and greater inhibitory control skills are more socially competent.

3.4.2. Externalizing behaviors

Parameter estimates for this model can be found in the middle panel of Table 4. Only the fixed effect of receptive vocabulary ($p < .05$) was significant. The parameter estimate suggests that children with greater receptive vocabulary have fewer externalizing behaviors.

3.4.3. Internalizing behaviors

Parameter estimates for this model can be found on the right panel of Table 4. The fixed effects of receptive vocabulary ($p < .01$) and inhibitory control (as measured by PT; $p < .01$) were significant. The parameter estimates suggest that children with greater receptive vocabulary and greater inhibitory control skills have fewer internalizing problems.

3.4.4. Moderation effects

The two-way interactions between inhibitory control and maternal education, maternal employment status, children's sex, ethnicity (Anglo-American vs. non Anglo-American), receptive vocabulary, emotional knowledge, and attention skills were each added separately to each of the above 2-level models to determine if the relationships between inhibitory control and social–emotional competence varied by these factors. None of the two-way interactions were significant in any of the three models. These findings suggested that the relationship between inhibitory control and social–emotional competence was similar for children with mothers of varying levels of education and employment status. It was also similar for boys and girls, Anglo-American and non-Anglo-American children and children with different levels of receptive vocabulary, emotional knowledge and attention skills.

4. Discussion

This study extends previous research by more closely examining the relationships between inhibitory control and social–emotional competence in low-income preschoolers. As hypothesized, the study found that children with better inhibitory control (as measured by PT) showed better behavioral adjustment compared to those with poorer inhibitory control skills. The findings indicated that after controlling for other variables known to be related to children's social–emotional competence, inhibitory control (as measured by PT) was a significant, unique predictor of social skills and internalizing problems. Children who had greater inhibitory control were more likely to be rated higher in social skills and lower in internalizing behavior. Although inhibitory control was not significant at the $p < .05$ level in the model predicting externalizing behavior, the parameter estimate indicated a relationship in the negative direction, which is consistent with the hypotheses. The exploratory moderation analyses indicated that these relationships did not vary by maternal education, maternal employment status or children's sex, ethnicity, receptive vocabulary, emotional knowledge or attention skills.

These findings provide further evidence for the central role of inhibitory control in distinguishing children with adjustment problems (Blair, 2003; Blair & Peters, 2003; Hughes et al., 1998, 2000). Research has consistently shown that inhibitory control is crucial in the development of social skills (Berlin & Bohlin, 2002; Brophy, Taylor, & Hughes, 2002; Hughes et al., 1998; Eisenberg, Spinrad, Fabes, Reiser, Cumberland, 2004) and may help explain the association between cognitive deficits and disruptive behavior

(Charman, Carroll, & Sturge, 2001; Quay, 1997; Rubia, Oosterlaan, Sergeant, Brandeis, & Leeuwen, 1998; Songuga-Barke, Dalen, Daley, & Remmington, 2002). In all, these findings suggest children's ability to inhibit prepotent responses is a significant predictor of social skills and problem behaviors above and beyond the impact of receptive vocabulary.

These conclusions must be tempered, however, by two considerations. First, like other measures of EF, PT and D/N are not pure measures of one individual component of EF. Instead, they measure inhibitory control in the context of a working memory demand. Although the results suggest that children's inhibitory control plays an important role in predicting children's social–emotional competence above and beyond maternal education and employment status, and children's sex, ethnicity, age, receptive vocabulary, emotional knowledge, and attention, the individual contribution of inhibitory control cannot be disentangled from working memory.

Second, the present study demonstrated that all inhibitory control tasks are not created equal. In the present analyses two commonly used measures of inhibitory control for young children, PT and D/N were only moderated correlated and were differentially related to social–emotional competence. Previous studies that utilized both of these measures created a composite inhibitory control score by averaging the PT and D/N scores (Blair, 2003; Blair & Peters, 2003), which masked the individual contributions of the two measures. Our analyses suggest that compared to D/N, PT is a more robust predictor of child behaviors. It may be that D/N is not as sensitive to individual differences in inhibitory control for disadvantaged preschool children or that the ability to inhibit a verbal response (as compared to a motor response) is less predictive of children's behavior. PT is designed to assess children's ability to inhibit a motor response, which particularly during early childhood when aggression is at its peak, may be more predictive of their social–emotional competence in the classroom.

4.1. Prevention implications

Numerous preschool prevention programs are aimed at improving children's intellectual ability (e.g., Ramey & Ramey, 2004; Weikart & Schweinhart, 1997). However, the results from the present study would suggest that inhibitory control skills are essential to children's social–emotional competence. Preschool children who have difficulties with inhibitory control skills are clearly at risk for lower social–emotional competence. Children who exhibit increased problem behaviors during early childhood are at risk for numerous poor developmental outcomes later in life including antisocial behavior, mental health problems, school failure, poor peer relationships, and unemployment (Broidy, Nagin, Tremblay, Bates, Brame, 2003; Caspi, Moffitt, Wright, & Silva, 1998; Henry, Caspi, Moffitt, & Silva, 1996; Jimerson, Egeland, Sroufe, & Carlson, 2000; Moffitt, 1993; Moffitt, Caspi, Dickson, Silva, & Stanton, 1996; Robins & Price, 1991; Tremblay, Phil, Vitaro, & Dobkin, 1994). Therefore, early identification of inhibitory control difficulties may be beneficial for targeting children at-risk for these maladaptive outcomes. Research highlights the role of environmental experience in the development of inhibitory control skills, suggesting there are many opportunities to intervene during early childhood (Dowsett & Livesey, 2000).

Although there are very few programs that directly target executive functions such as inhibitory control (Diamond, Barnett, Thomas, & Munro, 2007; Klingberg, Fernell, Olesen, Johnson, & Gustafsson, 2005; Rueda, Rothbart, McCandliss, Saccomanno, & Posner, 2005), researchers are increasingly recognizing the importance of incorporating theoretical models of neurocognitive functioning in the development and evaluation of prevention programs aimed at reducing problem behaviors and enhancing social competence (Greenberg, Riggs, & Blair, 2007; Riggs et al., 2006). Numerous evidence-based prevention programs whose goals are to promote self-regulation and reduce problem behaviors in preschoolers already currently exist (Domitrovich et al., 2007; Lynch, Geller, & Schmidt, 2004; Serna, Nielsen, Lambros & Forness, 2000; Webster-Stratton & Reid, 2003). Although these programs were not designed to enhance inhibitory control skills directly, it is possible that they influence them indirectly by targeting self-control and problem-solving skills.

4.2. Limitations

There are a number of limitations to the present study that should be considered when interpreting the results. First, due to the correlational and cross-sectional nature of our analyses it is not possible to determine the directionality of the associations between inhibitory control and children's social–emotional competence. A second limitation was the use of teacher ratings as the only assessment of social–emotional competence. Although teachers are an excellent source of information about children's behavior due to the extended periods of time they spend with them, biases may interfere with an accurate representation of their social–emotional competence. Future studies should include multiple measures from multiple sources in order to more accurately assess children's social–emotional competence.

A third concern regards the measurement issues that surround the assessment of inhibitory control and attention in young children (Blair, Zelazo, & Greenberg, 2005). No single, pure measure of inhibitory control in young children currently exists. As previously mentioned, PT and D/N are both tasks that require an ability to hold information in mind and therefore reflect children's working memory abilities in addition to their inhibitory control abilities. Inhibitory control is only one component of the important multi-dimensional construct of EF. Research has shown that other components of EF, including working memory and planning, are also important in the prediction of children's behavior (Espy, McDiarmid, Cwik, Stalets, Hamby, & Senn, 2004; Hughes et al., 1998; Hughes, Cutting, & Dunn, 2001; Moses, Carlson, & Sabbagh, 2005). Future studies should assess all aspects of EF to understand how different components of EF jointly or uniquely predict children's behavior. Similarly, this study used the Leiter-R AS subtest as our measure of attention. While it is a measure with strong psychometric properties, there are no studies that we are aware of that compare the Leiter-R AS, which requires a motor response to other measures of attention that do not.

Finally, one must be cautious when interpreting the lack of significant interaction effects as an indication that the relationship between inhibitory control and social–emotional competence did not vary by maternal education or employment status or children's sex, ethnicity, receptive vocabulary, emotional knowledge or attention skills. Although the interactions were entered one at a time to limit the effect of power issues, the sample size in the present study may have limited the ability to detect significant interactions. Future studies should replicate these moderation analyses with larger sample sizes to eliminate lack of power as the explanation for these findings.

4.3. Contributions and future directions

The present study makes significant contributions to the growing literature examining the relationship between neurocognitive skills and social–emotional competencies in young children. First, the majority of past research examining the link between inhibitory control and children's behavior has been conducted with relatively small, homogenous, middle-class samples. However, research suggests that inhibitory control skills are especially important sources of resilience for children raised in stressful environments (Blair, 2002). Therefore it is crucial to understand how inhibitory control operates in a more disadvantaged sample like the one in the present study. Second, unlike other studies that ignore the nested structure of their data, the present study used multi-level modeling in order to accurately account for classroom-level effects on individual-level relations between inhibitory control and teacher-rated social–emotional competence. Finally, few studies have isolated the specific impact of inhibitory control (in the context of a working memory demand) by covarying factors already known to be associated with social–emotional competence. The present study makes novel contributions by showing how children's performance on inhibitory control tasks are significantly associated with teacher ratings of classroom behaviors in a relatively large sample of high risk, ethnically diverse, low income preschoolers.

Despite the contributions made by the present study, there remain a number of areas of research that deserve further attention. The present study provided evidence of the significant role of inhibitory control to social–emotional competence, but future research should focus on examining the relationship between inhibitory control and other important developmental tasks in early childhood, including school readiness and academic achievement (Blair, 2002). Researchers have already begun to examine the associations between executive functions and academic skills. For instance, Espy et al. (2004) found that both working memory and inhibitory control contribute to the prediction of early math skills after controlling for child's age, maternal education and receptive vocabulary. Presumably, the development of these early neurocognitive abilities will lay the groundwork for the future self-regulatory and problem-solving skills needed to succeed both socially and academically throughout the life course.

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